



LIGHT -07

Class 10 - Science

Section A

1. A ray of light after refraction through a concave lens emerges parallel to the principal axis. Under what condition can it happen? [2]
2. A man stands in front of a mirror of Special shape. He finds that his image has a very small head, a fat body, and legs of normal size. What can we say about the shapes of three parts of the mirror? [2]
3. A lens immersed in a transparent liquid is not visible. Under what condition can this happen? [2]
4. You read a newspaper because of the light that it reflects. Then why do you not see even a faint image of yourself in the newspaper? [2]
5. A diver under water, looks obliquely at a fisherman standing on the bank of a lake. Would the fisherman look taller or shorter to the diver than what he actually is? [2]
6. Why do the trees and houses on the other side of an open ground appear to be shaking during the summer noon? [2]
7. Why is a concave mirror preferred to a plane mirror for Shaving? [2]
8. Plane and convex mirrors produce virtual images of objects. Can they produce real images under some circumstances? Explain. [2]
9. Will the reflected rays converge at a point when a parallel beam of light is incident on a concave mirror of large aperture? [2]
10. When does a convex lens behave as a concave lens? [2]
11. An air bubble in a jar of water shines brightly. Why? [2]
12. What type of lens is an air bubble inside water? Give the reason also. [2]
13. An object of height 4.0 cm is placed at a distance of 30 cm from the optical centre O of a convex lens of focal length 20 cm. Draw a ray diagram to find the position and size of the image formed. Mark optical centre O and principal focus F on the diagram. Also, find the approximate ratio of the size of the image to the size of the object. [2]
14. The wall of a room is covered with a perfect plane mirror, and two movie films are made, one recording the movement of a man and the other of his mirror image. From viewing the films later, can an outsider tell which is the mirror image film? [2]
15. Cinema screens are made rough and white. why? [2]
16. Why a concave mirror of small aperture forms a sharper image? [2]

Section B

17. **Assertion (A):** When the object moves with a velocity $\vec{v}$, its image in the plane mirror moves with a velocity of $-2\vec{v}$. [1]

Reason (R): The minimum height of the mirror to be required to see the full image of man of height h is $\frac{h}{2}$.

- a) Both A and R are true and R is the correct b) Both A and R are true but R is not the

- explanation of A. correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
18. **Assertion (A):** As the temperature of a medium increases the refractive index decreases. [1]
Reason (R): When a ray travels from a vacuum to a medium, then μ is known as the absolute refractive index of the medium. ($\mu_{\text{vacuum}} = 1$).
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
19. **Assertion (A):** The refractive index of diamond is $\sqrt{6}$ and refractive index of liquid is $\sqrt{3}$. If the light travels [1]
from the diamond to the liquid, it will initially be reflected when the angle of incidence is 30° .
Reason (R): $\mu = \frac{1}{\sin C}$, where μ is the refractive index of diamond with respect to liquid.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
20. **Assertion (A):** Red light travels faster in glass than green light. [1]
Reason (R): The refractive index of glass is less for red light than for green light.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
21. **Assertion (A):** As light travels from one medium to another, the frequency of light does not change. [1]
Reason (R): Because frequency is the characteristic of source.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
22. **Assertion (A):** A plane mirror neither converges parallel rays of light nor diverges them. [1]
Reason (R): The focal length of a plane mirror can be considered to be infinite.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
23. **Assertion (A):** If a spherical mirror is dipped in water, its focal length remains unchanged. [1]
Reason (R): A laser light is focused by a converging lens. There will be a significant chromatic aberration.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
24. **Assertion (A):** Linear magnification of a mirror has no unit. [1]
Reason (R): The ratio of height of the image to the height of the object is the linear magnification produced by mirror.

- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
25. **Assertion (A):** Convex mirrors are used for rear view on vehicles. [1]
Reason (R): The size of the image formed by a convex mirror will be same.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
26. **Assertion(A):** Concave mirrors are used as make-up mirrors. [1]
Reason (R): When the face is held within the focus of a concave mirror, then a diminished image of the face is seen in the concave mirror.
- a) Both A and R are true and R is the correct explanation of the assertion. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
27. **Assertion (A):** The speed of light in glass depends upon the colour of light. [1]
Reason (R): The speed of light in glass $v_g = \frac{c}{n_g}$, the refractive index of glass is different for different colours.
- a) Both A and R are true and R is the correct explanation of the assertion. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
28. **Assertion (A):** Keeping a point object fixed, if a plane mirror is moved, the image will also move. [1]
Reason (R): In the case of a plane mirror, distance of object and Its image is equal from any point on the mirror.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
29. **Assertion (A):** Light bends from its path when it goes from one medium to another medium. [1]
Reason (R): Speed of light changes, when it goes from one medium to another medium.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
30. **Assertion (A):** A concave lens of very short focal length causes higher divergence than one with longer focal length. [1]
Reason (R): The power of a lens is directly proportional to its focal length.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
31. **Assertion (A):** A ray of light that travels obliquely from one transparent medium into another will change its direction in the second medium. [1]

Reason (R): Refraction is due to change in the speed of light as it enters from one transparent medium to another.

- a) Both A and R are true and R is the correct explanation of A.
- b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false.
- d) A is false but R is true.

32. **Assertion (A):** Mirror formula can be applied to a plane mirror. [1]

Reason (R): A plane mirror is a spherical mirror of infinite focal length.

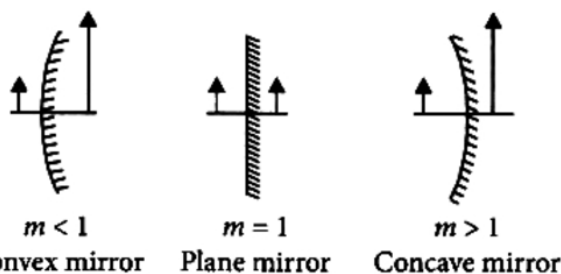
- a) Both A and R are true and R is the correct explanation of the assertion.
- b) Both A and R are true but R is not the correct explanation of the assertion.
- c) A is false but R is true.
- d) A is true but R is false.

33. **Assertion (A):** If both the object and plane mirror are moved through a distance of x then the image moves through a distance of $2x$. [1]

Reason (R): If the object is fixed and plane mirror is moved through a distance x then the image also moves through a distance of $2x$.

- a) Both A and R are true and R is the correct explanation of the assertion.
- b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false.
- d) A is false but R is true.

34. **Assertion (A):** We can decide the nature of a mirror by observing the size of erect image in the mirror. [1]



Reason (R): The minimum distance between a real object and its real image in a concave mirror is non zero.

- a) Both A and R are true and R is the correct explanation of A.
- b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false.
- d) A is false but R is true.

35. **Assertion (A):** There exist two angles of incidence for the same magnitude of deviation (except minimum deviation) by a prism kept in the air. [1]

Reason (R): In a prism kept in air, a ray is incident on first surface and emerges out of second surface. Now, if another ray is incident on second surface (or prism) along the previous emergent ray, then this ray emerges out of first surface along the previous incident ray. This phenomenon is called the principle of reversibility of light.

- a) Both A and R are true and R is the correct explanation of A.
- b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false.
- d) A is false but R is true.

36. **Assertion (A):** A ray incident along normal to the mirror retraces its path. [1]

Reason (R): In reflection, the angle of incidence is always equal to the angle of reflection.

- a) Both A and R are true and R is the correct
- b) Both A and R are true but R is not the

- explanation of A. correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
37. **Assertion (A):** Light travels faster in air than in glass. [1]
Reason (R): Air is denser than glass.
- a) Both A and R are true and R is correct explanation of the assertion. b) Both A and R are true but R is not the correct explanation of the assertion.
- c) A is false but R is true. d) A is true but R is false.
38. **Assertion (A):** Higher is the refractive index of a medium or denser the medium, lesser is the velocity of light in that medium. [1]
Reason (R): Refractive index is inversely proportional to velocity.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
39. **Assertion (A):** In the case of concave mirror, the minimum distance between real object and its real image is zero. [1]
Reason (R): If concave mirror forms virtual image of real object, the image is magnified.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
40. **Assertion (A):** Refractive index of glass with respect to air is different for red light and violet light. [1]
Reason (R): Refractive index of a pair of media depends on the wavelength of light used.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
41. **Assertion (A):** A convex mirror is used as a driver's mirror. [1]
Reason (R): Because convex mirror's field of view is large and images formed are virtual, erect and diminished.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
42. **Assertion (A):** It is impossible to see a virtual image. [1]
Reason (R): The rays that seem to emanate from a virtual image don't, in fact, emanate from the image.
- a) Both A and R are true and R is the correct explanation of the assertion. b) Both A and R are true but R is not the correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
43. **Assertion (A):** The height of an object is always considered positive. [1]
Reason (R): An object is always placed above the principal axis in the upward direction.
- a) Both A and R are true and R is the correct explanation of the assertion. b) Both A and R are true but R is not the correct explanation of A.

- c) A is true but R is false. d) A is false but R is true.
44. **Assertion (A):** When the object moves with a velocity of 2 m/s, its image in the plane mirror moves with the velocity of 4m/s. [1]
Reason (R): The image formed by the plane mirror is as far behind the mirror as the object is in front of it.
- a) Both A and R are true and R is the correct explanation of the assertion. b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false. d) A is false but R is true.
45. **Assertion (A):** Speed of light in glass of $\mu = 1.5$ is 2×10^8 m/sec [1]
Reason (R): According to dual theory, light has particle nature and wave nature simultaneously.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false. d) A is false but R is true.
46. **Assertion (A):** A plane mirror may form a real image. [1]
Reason (R): Plane mirror forms a virtual image if the object is real.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false. d) A is false but R is true.
47. **Assertion (A):** For observing traffic at back, the driver mirror is a convex mirror. [1]
Reason (R): A convex mirror has a much larger field of view than a plane mirror.
- a) Both A and R are true and R is the correct explanation of the assertion. b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false. d) A is false but R is true.
48. **Assertion (A):** Property of converging of a convergent lens does not remain same in all media. [1]
Reason (R): Property of lens whether the ray is diverging or converging is independent of the surrounding medium.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false. d) A is false but R is true.
49. **Assertion (A):** When a concave mirror is held underwater, its focal length will increase. [1]
Reason (R): The focal length of a concave mirror is independent of the medium in which it is placed.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false. d) A is false but R is true.
50. **Assertion (A):** If the rays are diverging after emerging from a lens; the lens must be concave. [1]
Reason (R): The convex lens can give diverging rays.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.

- c) A is true but R is false. d) A is false but R is true.
51. **Assertion (A):** The refractive index of Kerosene is 1.44 which is optically denser than water. [1]
Reason (R): The mass density of kerosene is lesser than water.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false. d) A is false but R is true.
52. **Assertion (A):** A virtual image cannot be photographed. [1]
Reason (R): Only real objects are photographed.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false. d) A is false but R is true.
53. **Assertion (A):** If a ray of light is incident on a convex mirror along its principal axis, then the angle of incidence, as well as the angle of reflection for a ray of light, will be zero. [1]
Reason (R): A ray of light going towards the centre of curvature of a convex mirror is reflected back along the same path.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false. d) A is false but R is true.
54. **Assertion (A):** The size of the mirror affect the nature of the image. [1]
Reason (R): Small mirrors always form virtual images.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false. d) A is false but R is true.
55. **Assertion (A):** The small object, to be seen in a microscope, is kept within the two foci of its objective. [1]
Reason (R): In this case, the image formed by the objective is nearer to the eyepiece.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false. d) A is false but R is true.
56. **Assertion (A):** In diffused reflection, a parallel beam of incident light is reflected in different direction. [1]
Reason (R): The diffused reflection of light is due to the failure of the laws of reflection.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false. d) A is false but R is true.
57. **Assertion (A):** Propagation of light through an optical fibre is due to total internal reflection taking palce at the core-clade interface. [1]
Reason (R): Refractive index of the material of the core of the optical fibre is greater than that of air.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.

- c) A is true but R is false. d) A is false but R is true.
58. **Assertion (A):** A point object is placed at a distance of 26 cm from a convex mirror of focal length 26cm. The image will not form at infinity. [1]
Reason (R): For above-given system the equation $\frac{1}{u} + \frac{1}{v} = \frac{1}{f}$ gives $v = \infty$
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false. d) A is false but R is true.
59. **Assertion (A):** A ray of the light incident along the normal to the plane mirror retraces its path after reflection from the mirror. [1]
Reason (R): A ray of light along the normal has an angle of incidence as $\frac{\pi}{2}$ and hence, it retraces its own path after reflection from the mirror.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false. d) A is false but R is true.
60. **Assertion (A):** Light is able to reach earth from the sun. [1]
Reason (R): Light rays can travel in vacuum.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false. d) A is false but R is true.
61. **Assertion (A):** Large concave mirrors are used to concentrate sunlight to produce heat in solar cookers. [1]
Reason (R): Concave mirror converges the light rays falling on it to a point.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false. d) A is false but R is true.
62. **Assertion (A):** Light rays retrace their path when their direction is reversed (Law of reversibility of light rays). [1]
Reason (R): For the refraction light, water is denser than air, but for the refraction of sound, water is rarer than air.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false. d) A is false but R is true.
63. **Assertion (A):** The image formed by a concave mirror is certainly real if the object is virtual. [1]
Reason (R): The image formed by a concave mirror is certainly virtual if the object is real.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false. d) A is false but R is true.
64. **Assertion (A):** The SI unit of power of lens is **diopetre**. [1]
Reason (R): The power of a concave lens is positive and that of a convex lens is negative.
- a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.

- explanation of A. correct explanation of A.
- c) A is true but R is false. d) A is false but R is true.
65. **Assertion (A):** The image of a virtual object formed by a thin converging lens is always real. [1]
Reason (R): In the case of a thin lens, $\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$.
 a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
 c) A is true but R is false. d) A is false but R is true.
66. **Assertion (A):** A convex lens is made of two different materials. A point object is placed on the principal axis. [1]
 The number of images formed by the lens will be two.
Reason (R): The image formed by convex lens is always virtual.
 a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
 c) A is true but R is false. d) A is false but R is true.
67. **Assertion (A):** The focal length of the convex mirror will increase if the mirror is placed in water. [1]
Reason (R): The focal length of a convex mirror of radius R is equal to, $f = \frac{R}{2}$
 a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
 c) A is true but R is false. d) A is false but R is true.
68. **Assertion (A):** The nature of the image formed by a concave mirror is real and inverted. [1]
Reason (R): The image formed by a concave mirror depends on the position of the object at infinity.
 a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
 c) A is true but R is false. d) A is false but R is true.
69. **Assertion (A):** An object is placed at a distance of f from a convex mirror of focal length f, its image will form at infinity. [1]
Reason (R): The distance of an image in a convex mirror can never be infinity.
 a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.
 c) A is true but R is false. d) A is false but R is true.
70. **Assertion (A):** The mirrors used in searchlights are concave spherical. [1]
Reason (R): In the concave spherical mirror the image formed is always virtual.
 a) Both A and R are true and R is the correct explanation of the assertion. b) Both A and R are true but R is not the correct explanation of A.
 c) A is true but R is false. d) A is false but R is true.
71. **Assertion (A):** Refractive index has no units. [1]
Reason (R): The refractive index is a ratio of two similar quantities.
 a) Both A and R are true and R is the correct explanation of A. b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.